

NEONATAL JAUNDICE

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NNJ PHOTOTHERAPY SB THRESHOLDS							
Age in Days	Age in Hours of Life	TcB Levels to trigger SB		SB Levels for Phototherapy		Critical (Emergency) SB Levels	
		Normal Risk TcB	High Risk TcB	Normal Risk SB	High Risk SB	Normal Risk SB	High Risk SB
Day 1	0 to 24	Managed in hospital					
Day 2	25 to 36	160	140	200	175	300	275
	37 to 48	180	160	225	200	350	300
Day 3	49 to 72	200	180	250	225	350	325
Day 4	73 to 96	220	200	275	250	375	350
Day 5	97 to 120	220	200	275	250	400	350
Day 6 to Day 7	121 to 168	240	220	300	275	400	375
Day 8 to Day 14		250	240	325	300	425	375
Day 15 to Day 30				325	300	425	375
		Management should be based on SB		For Direct Admission to NUH or KKH (To reach within 4 hours)		Refer to NUH/KKH CE (To reach within 1 hour)	

Normal Risk - Gestational age ≥ 37 weeks - No high-risk factors <u>Abbreviations</u> TcB Transcutaneous Bilirubinometry SB Serum Bilirubin. Ordered as "Bilirubin, Paeds" CE Children's Emergency	High Risk Factors - Jaundice observed in first 24 hrs - Late Prematurity (35 to 36⁺⁶ weeks) - Asphyxia (Apgar ≤ 5 at 1 and 5 min) - Term IUGR with birth weight 2000 - 2500g - Family history of severe NNJ in siblings needing exchange transfusion - G6PD deficiency and other haemolytic conditions *ABO incompatibility defined as - Mother's blood group O and baby's blood group is A/B AND - DCT positive OR maternal "Anti-A" or "Anti-B" IgG antibodies titre ≥ 128 - Rhesus incompatibility - Rapid rate of rise of SB > 103µmol/L in 24 hours *Exclusive breastfeeding AND weight loss ≥ 10% of birth weight	Not for Management in Primary Care Setting Neonates <35 weeks gestation - To check SB against the preemie chart at the respective nursery to evaluate on threshold for Phototherapy and refer to the hospitals for the next review. - NUH Nursery hotline: 67725480 / 67723480 - KKH Nursery hotline: 63941882 - Please raise above cases to Head / Dep to review. - or Birth weight <2000g are not to be managed in the primary care setting.
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Updated 11/2019 based on National NNJ Practice Guidelines 2018. Please refer to National PG for more details and [Intranet](#) for updates.

- * NB: If mother's group & antibody titres unknown, manage the baby as 'High Risk' until the results are available
- * The following group of infants from the second week of life onwards (8-30 days of age) could be considered as normal risk
- Mother's group & antibody titres still unknown
 - Exclusive breastfeeding who had weight loss ≥ 10% & regained normal weight

APPROACH TO A JAUNDICED NEONATE

Age up to 14 days

Objective: Risk stratify and catch peak jaundice level

FIRST VISIT	
1. TcB or SB	Nurse will do TcB or send for SB (if TcB contraindicated or crosses phototherapy thresholds in the high-risk category.) Nurse will transfer to Dr for First Visit Assessment . See nursing TcB workflow here.
2. Neonate's Age	Calculate age of neonate in hours of life at time of TcB/SB (If neonate's age just falls between age categories, follow next category. E.g. 48h 10min follows "49-72h" category)
3. Initial Risk Assessment	Check for high risk factors as listed in chart above.
4. Current Clinical Status	Feeding – ask type of feeding and if baby is feeding well Check for pale stools and document. (Use stool chart.) Assess hydration status and percentage of weight loss.
5. Physical examination	General wellbeing – exclude unwell child, dehydration Note level of jaundice, exclude gross organomegaly.
6. Document Risk assessment	State risk assessment clearly in notes to facilitate nurses TcB interpretation in subsequent visits.
7. Doctor management	Any unwell/sick baby with NNJ should be referred to CE immediately. Review SB and check rate of rise of SB. Refer if SB at admitting phototherapy level in the correct risk category . Initiate Direct Admission if eligible (see box on right). If not for phototherapy, and baby is ≤ 14 days old, order Nurse TcB.
8. Educate Parents	See section on next page

REPEAT VISITS	
1. TcB or SB*	Nurse will do TcB or send for SB according to nursing workflow. Nurse will refer to Doctor if i. SB crosses phototherapy threshold OR ii. Up-trending TcB/SB level and expected TCU is on clinic closure date If SB expected to rise to phototherapy level on clinic closure date, NUH: Refer to CE as per usual. KKH: Refer to NNJ Sunday/PH Clinic M with PIL, by appointment
* Refer to <i>Annex A</i> for workflow	
2. Reassess Clinical Status	Reassess clinical status as this may change risk category of the neonate - feeding, stool colour and percentage of weight loss Check that child is well, and note level of jaundice.
3. Doctor management	Any unwell/sick baby should be referred to CE immediately. Review SB and check rate of rise of SB. Check if any change in risk category before referring for phototherapy. Initiate Direct Admission if eligible (see box on right).

INDICATIONS FOR CHILDREN'S EMERGENCY (Contraindications to Direct Admission)
Infants with SB at admitting phototherapy level but i. Critical (Emergency) SB levels as shown in the chart ii. To reach within 1 hour iii. Baby unwell - Fever, poor suck, clinical evidence of dehydration. Call ambulance if signs of encephalopathy – hypo/hypertonia, excessive drowsiness, lethargy. iv. Positive maternal contact history with chickenpox.

INDICATIONS FOR DIRECT ADMISSION
i. SB at admitting phototherapy level (refer to chart) and age up to 14 days old. ii. No contraindications to the above (Children's CE) iii. Parents agreeable for Direct Admission iv. To reach within 4 hours
Communication is important - current SB level warrants further observation and management as inpatient in Neonatology. Specialist will make the final decision as to whether phototherapy is needed. They will repeat SB and sometimes it would have started to drop below phototherapy level.
Workflows for Direct Admissions i. All DA referrals to be endorsed by a Senior Doctor before proceeding. ii. NUH Neonatology: Create REFERRAL TO NEONATOLOGY – NUH (DIRECT ADMISSION) (To loc/pos: "No preference") and direct parents to referral counter. iii. KKH: Create e-referral, fill up manual form and direct parents to KKH CE Admission Counter . Indicate "Direct Admission for Phototherapy" in e-referral.

CRITERIA FOR DISCHARGE
i. Look for downward trend. ii. Clinically not jaundiced, and no pale stools. iii. Monitor until TcB < 80µmol/L or SB < 100µmol/L . iv. Parents to continue to monitor jaundice and stool colour.

Age > 14 days

Objective: Exclude conjugated hyperbilirubinaemia

REPEAT VISITS	
1. Serum Bilirubin (SB)	After 14 days old, order “ Bilirubin, Paeds ” which uses heel prick.
2. Reassess Clinical Status	Feeding - ask type of feeding and if baby is feeding well Check for pale stools and document. Assess hydration and current weight. Check that child is well and regained birth weight . Note level of jaundice.
3. Doctor management	Any unwell/sick baby with NNJ should be referred to CE immediately. If SB > 325 µmol/L for Normal Risk neonates, and > 300 µmol/L for High-Risk neonates, refer to CE (not for direct admission.) Order a “Bilirubin, Direct” <u>Day 21 onwards</u> if SB still >100 µmol/L. Reinforce parental education.

Age > 30 days

Objective: Identify prolonged jaundice

REPEAT VISITS	
1. Total Bilirubin (TB)	After 30 days old, order “ Bilirubin, Total ” (TB). (SB not validated for use in babies > 30 days old.)
2. Reassess Clinical Status	Check feeding, stool colour and current weight. Check that child is well and ensure weight gain . Inspect stool specimen. Note level of jaundice.
3. Doctor management	If TB still > 100 µmol/L, clinically still jaundiced, ensure • Direct bilirubin done and < 20 µmol/L • No acholic stools Refer to Paeds Med after 4 weeks old.

Advice to Parents

1. Explain why baby is jaundiced. Stress importance of close monitoring. If jaundice too severe, it can damage baby's brain.
2. Sunning the baby is ineffective and may cause sunburn/ dehydration.
3. There is usually no indication to stop breast-feeding. Breast milk is best for the baby. Ensure adequate breastfeeding to keep baby well hydrated.
4. Some herbs increase risk of neurotoxicity and should be avoided during pregnancy/first month of breastfeeding: 1) Chuen-Lin (Chuan Lian), 2) Ngan-Huang (Niu Huang), 3) Yin-Chen
5. When ordering direct bilirubin, explain that extra blood will be drawn to distinguish the TYPE of jaundice. It is NOT a “liver test”.
6. Continue to monitor jaundice and stool colour on discharge. Bring baby back if yellowing recurs/persists or stools turn pale.

INDICATIONS FOR REFERRAL TO PAEDS MEDICINE

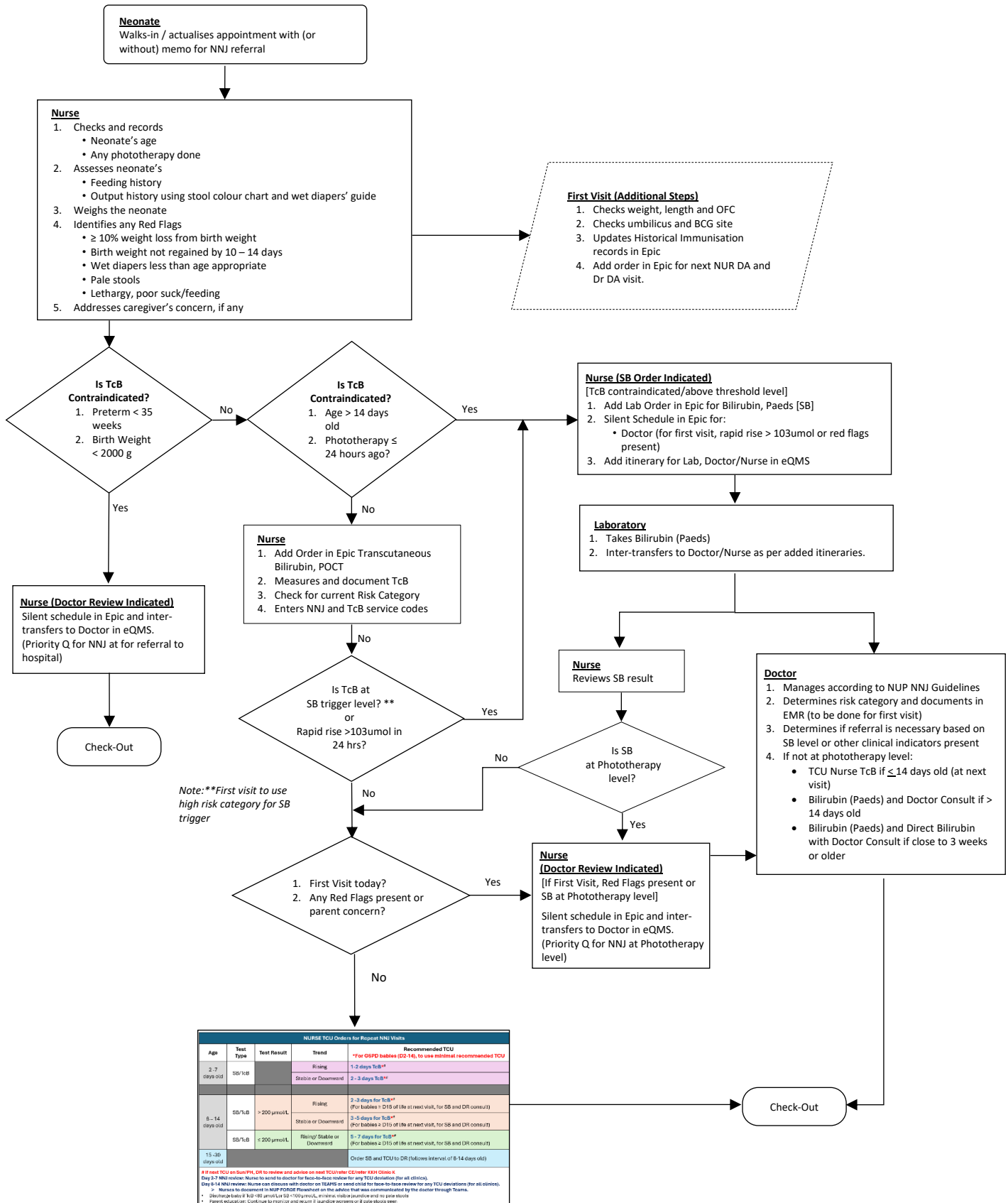
1. Urgent referral (**within one week**) if any evidence of conjugated hyperbilirubinaemia:
 - **Pale stools or**
 - **Direct bilirubin $\geq 20 \mu\text{mol/l}$**
2. Prolonged jaundice after 4 weeks old

CRITERIA FOR DISCHARGE

Ensure all of the following:

- i. **Direct bilirubin < 20 µmol/L**
- ii. Look for downward trend.
- iii. Clinically not jaundiced, and no pale stools.
- iv. Monitor until **SB/TB < 100 µmol/L**
- v. Parents to continue to monitor jaundice and stool colour

ANNEX A: NURSING WORKFLOW FOR NEONATAL JAUNDICE ASSESSMENT (PHASE 4)



ANNEX B: NURSE TCU ORDERS FOR REPEAT NNJ VISITS

NURSE TCU Orders for Repeat NNJ Visits				
Age	Test Type	Test Result	Trend	Recommended TCU <i>*For G6PD babies (D2-14), to use minimal recommended TCU</i>
2 -7 days old	SB/TcB		Rising	1-2 days TcB*#
			Stable or Downward	2 - 3 days TcB*#
8 – 14 days old	SB/TcB	> 200 µmol/L	Rising	2 -3 days for TcB*# (For babies ≥ D15 of life at next visit, for SB and DR consult)
			Stable or Downward	3 -5 days for TcB*# (For babies ≥ D15 of life at next visit, for SB and DR consult)
	SB/TcB	≤ 200 µmol/L	Rising / Stable or Downward	5 - 7 days for TcB*# (For babies ≥ D15 of life at next visit, for SB and DR consult)
15 -30 days old				Order SB and TCU to DR (follows interval of 8-14 days old)
# If next TCU on Sun/PH, DR to review and advice on next TCU/refer CE/refer KKH Clinic K Day 2-7 NNJ review: Nurse to send to doctor for face-to-face review for any TCU deviation (for all clinics). Day 8-14 NNJ review: Nurse can discuss with doctor on TEAMS or send child for face-to-face review for any TCU deviations (for all clinics). ➤ Nurses to document in NUP FORGE Flowsheet on the advice that was communicated by the doctor through Teams. - Discharge baby if TcB <80 µmol/Lor SB <100 µmol/L, minimal visible jaundice and no pale stools - Parent education: Continue to monitor and return if jaundice worsens or if pale stools seen				